

- **Unit Testing**

As the name suggests, the unit test isolates each unit of the software and tests individual components/modules against the pre-defined specification and requirements. It is also known as a module or component testing. It is performed by developers when the software is getting developed to verify whether each module of the system works as designed. It is less expensive as defects can be found in the early stages and can be fixed.

- **Integration Testing**

Integration testing is an extension of the unit test and is usually performed after unit testing is complete. The software contains multiple modules that communicate with each other. Integration testing combines all these individual modules and tests them together to check the connectivity between modules. It is further broken into a top-down or bottom-up approach. Bottom-up integration testing begins with unit testing which is later followed by testing of higher-level modules. Unlike bottom-up testing, Top-down integration testing tests high-level modules first followed by lower-level modules.

- **System Testing**

System testing is the testing of the entire software as one complete unit to verify if the software meets the specified requirements and quality standards. It is the most important testing among all as it tests the software in an environment which is very close to the production environment and validates the user's interaction experience with the software.

- **Acceptance Testing**

This is the last level of functional software testing which is performed before making the software available to the end-users. As it is obvious from the name, acceptance testing is done to take customer sign-off so that the software can be released in the market. The main goal of this testing is to evaluate if the software complies with the business specifications, meets the quality standards, and is ready to be delivered to the end-users.

2. Non-functional Testing

Non-functional testing tests the software for non-functional attributes such as load, performance, speed, stability, scalability, etc. It tests the software for requirements that are non-functional but important. It has the following types: